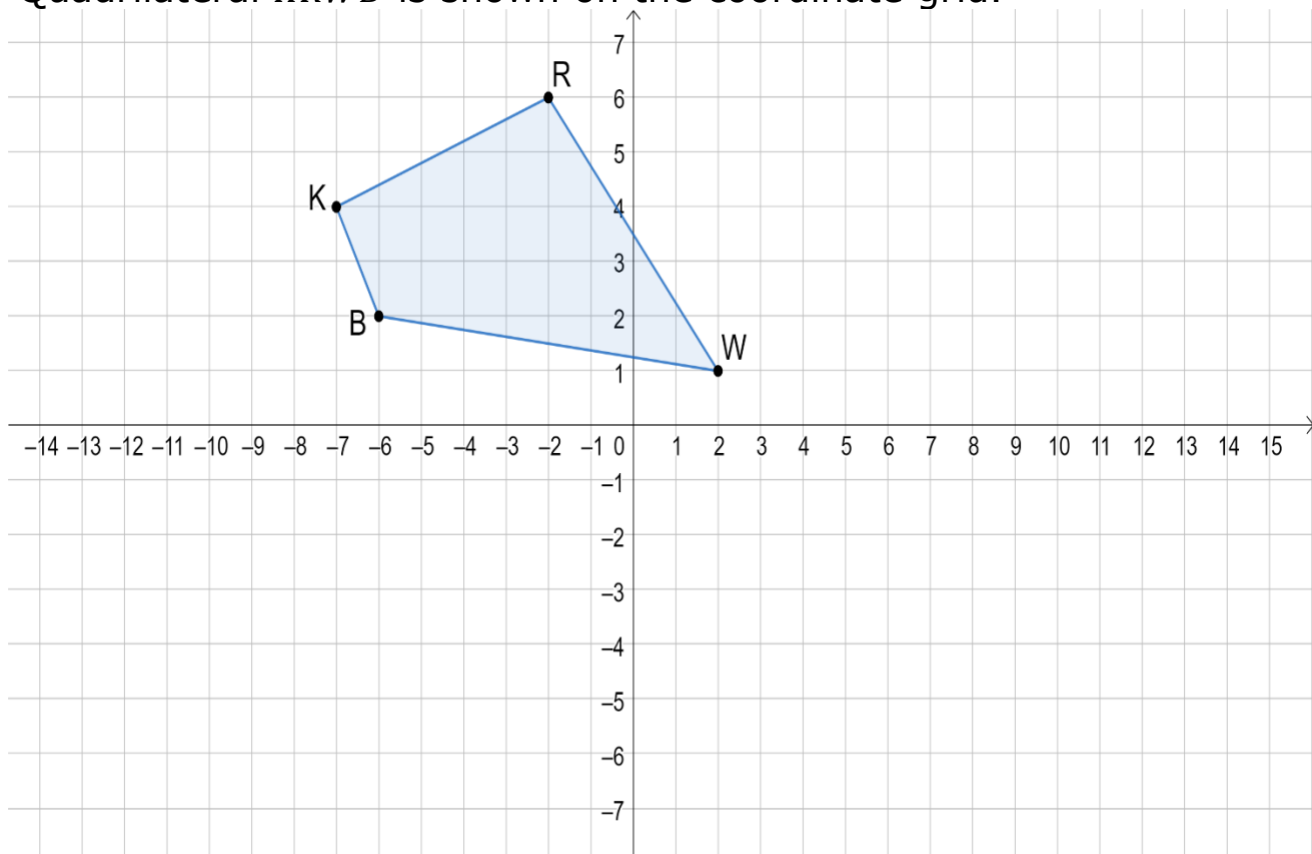
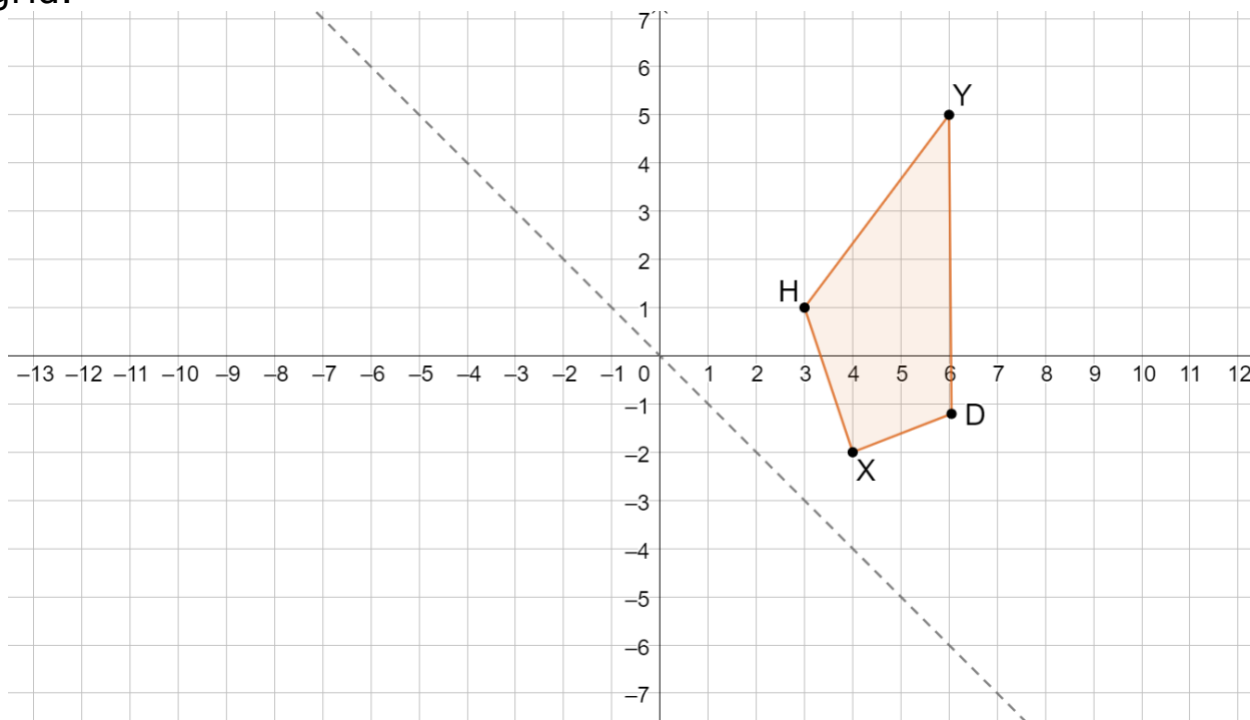


1. Quadrilateral $KRWB$ is shown on the coordinate grid.



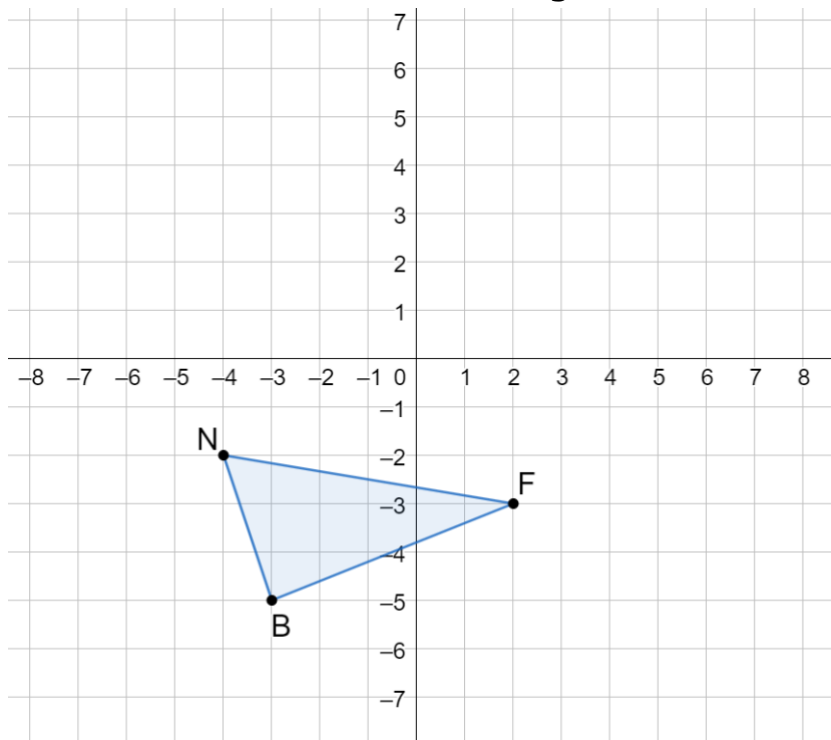
- a. Apply the sequence of transformations to quadrilateral $KRWB$. Draw the result of each transformation on the coordinate grid.
- $(x, y) \rightarrow (x, -y)$ Label as $K'R'W'B'$.
 - $(x, y) \rightarrow (x + 10, y + 6)$ Label as $K''R''W''B''$.
- b. Explain how the sequence of transformations for quadrilateral $KRWB$ resulted in $KRWB \cong K''R''W''B''$.

2. Quadrilateral $YDXH$ and the line $y = -x$ are shown on the coordinate grid.



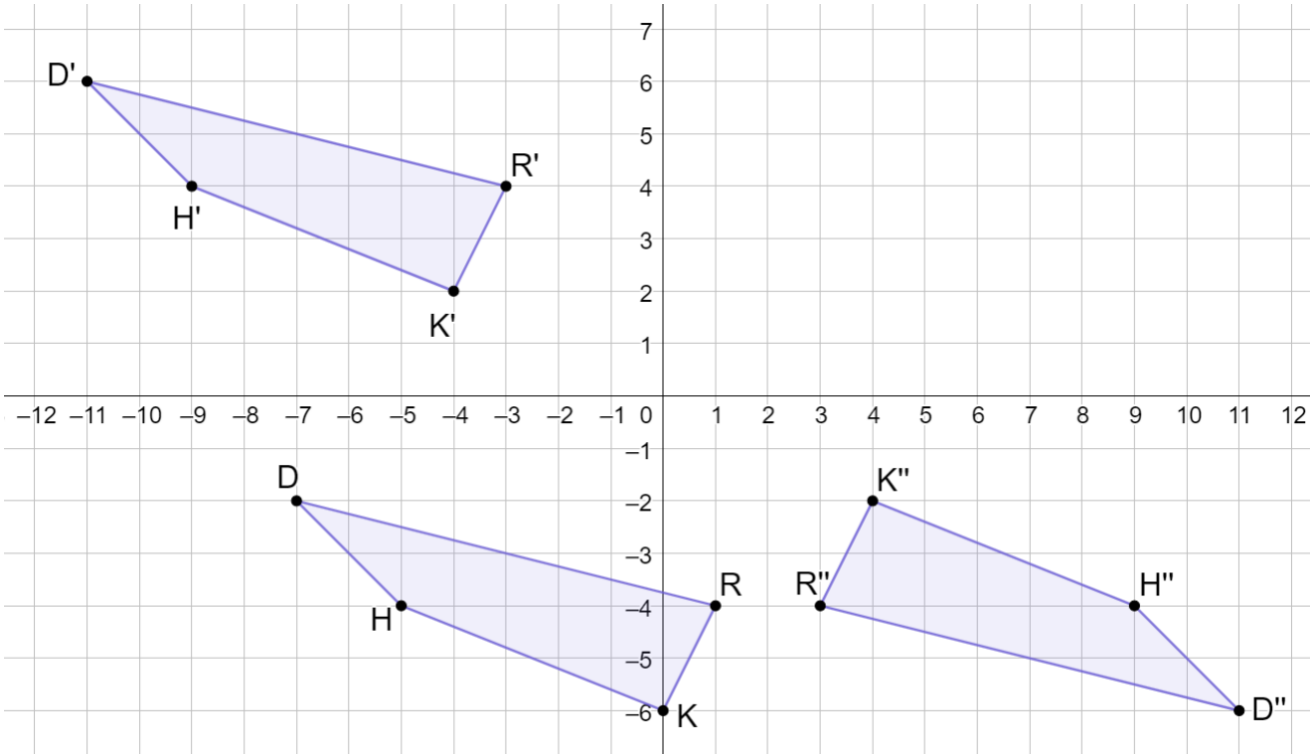
Reflect quadrilateral $YDXH$ across the line $y = -x$. Draw the result of the reflection on the coordinate grid.

3. Triangle BFN is shown on the coordinate grid.



Rotate $\triangle BFN$ 90° counterclockwise about the origin. Draw the result of the rotation on the coordinate grid.

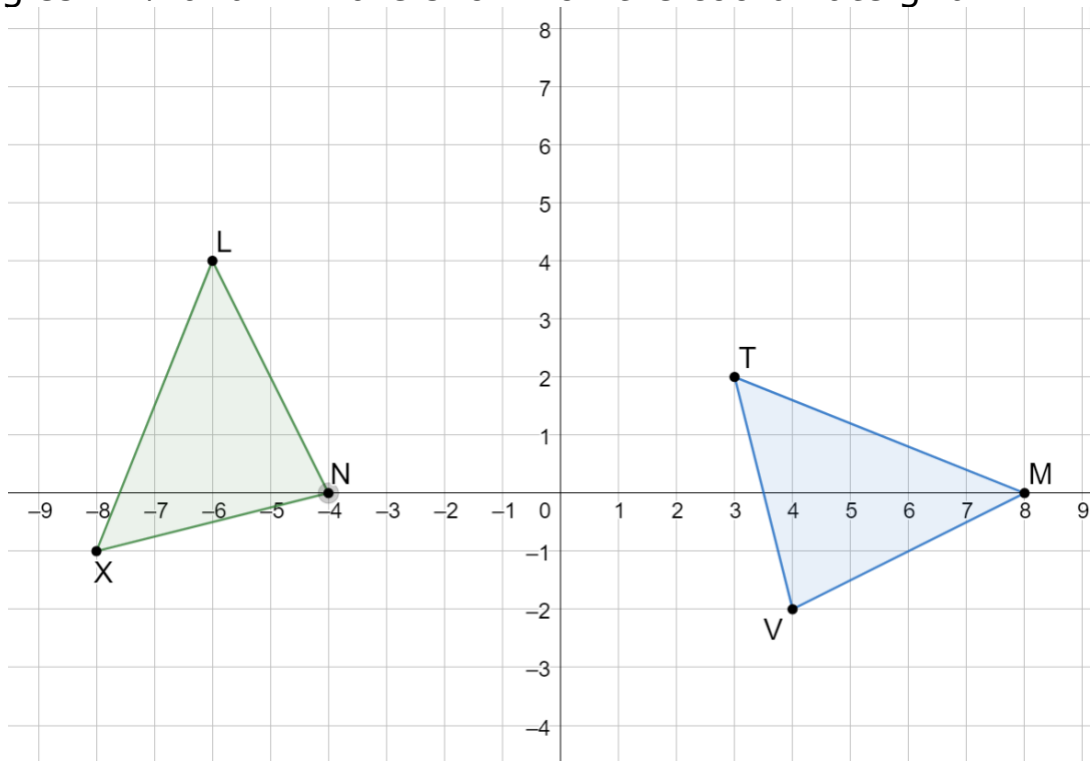
4. Quadrilateral $RKHD$ was transformed to create quadrilateral $R'K'H'D'$. Then, quadrilateral $R'K'H'D'$ was transformed to create quadrilateral $R''K''H''D''$. The quadrilaterals are shown on the coordinate grid.



Complete the table by writing a description for each of the transformations.

	Quadrilateral $RKHD$ Transformed to Quadrilateral $R'K'H'D'$	Quadrilateral $R'K'H'D'$ Transformed to Quadrilateral $R''K''H''D''$
Written Description		

5. Triangles TMV and LNX are shown on the coordinate grid.



- a. Complete the statement.

The sequence of transformations that will map $\triangle TMV$ onto $\triangle LNX$

is first

- translate $\triangle TMV$ left 11, up 1,
- reflect $\triangle TMV$ across the y -axis,
- rotate $\triangle TMV$ 90° clockwise about the origin,
- rotate $\triangle TMV$ 90° counterclockwise about the origin,

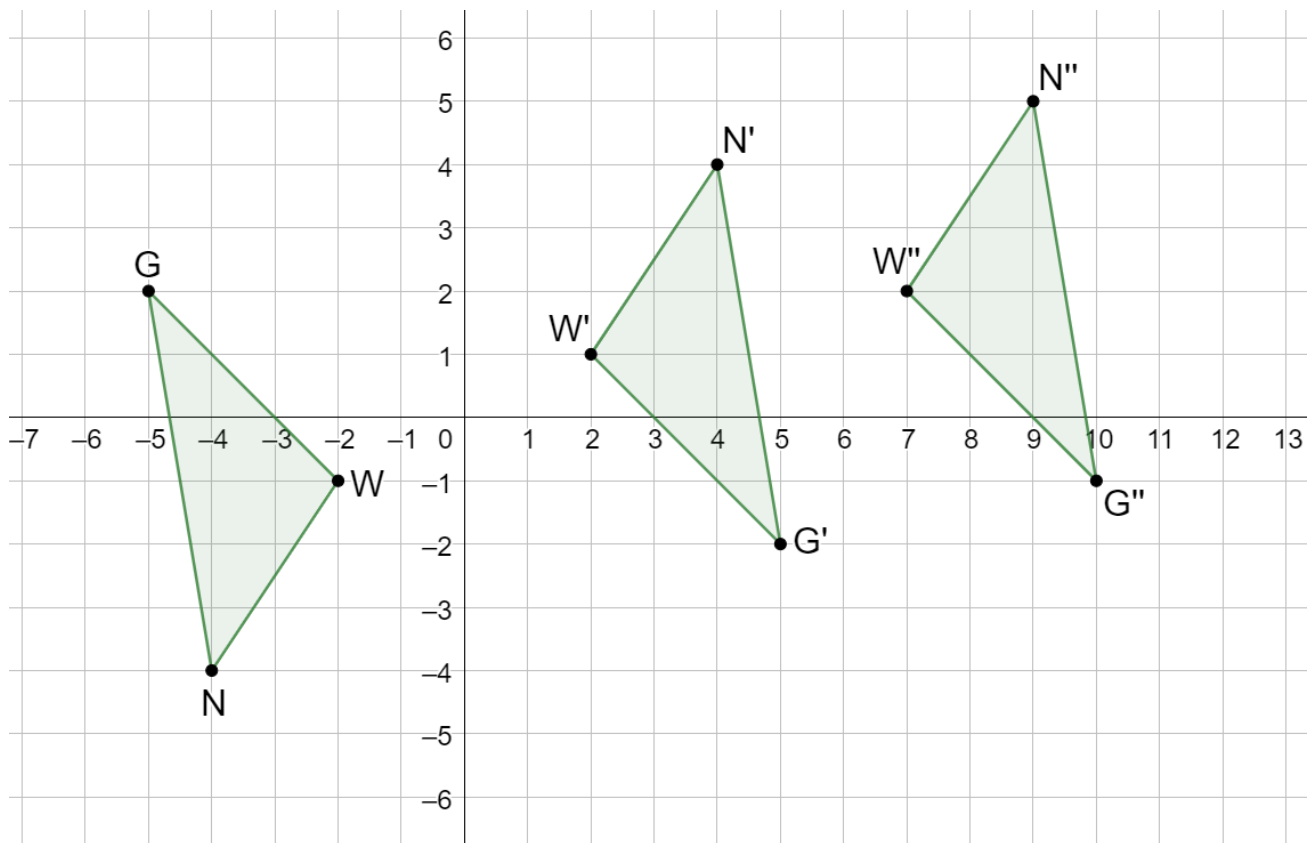
then

- translate $\triangle T'M'V'$ left 3, up 2.
- translate $\triangle T'M'V'$ left 2, down 3.
- translate $\triangle T'M'V'$ left 6, down 4.
- reflect $\triangle T'M'V'$ across the $y = -2$.
- rotate $\triangle TMV$ 90° counterclockwise about the origin.

- b. Complete the statements by filling in the blanks.

Vertex M maps to vertex ____, vertex T maps to vertex ____, and vertex V maps to vertex _____. Therefore, $\triangle TMV$ is congruent to $\triangle$ ____ by the definition of congruence in terms of rigid motions.

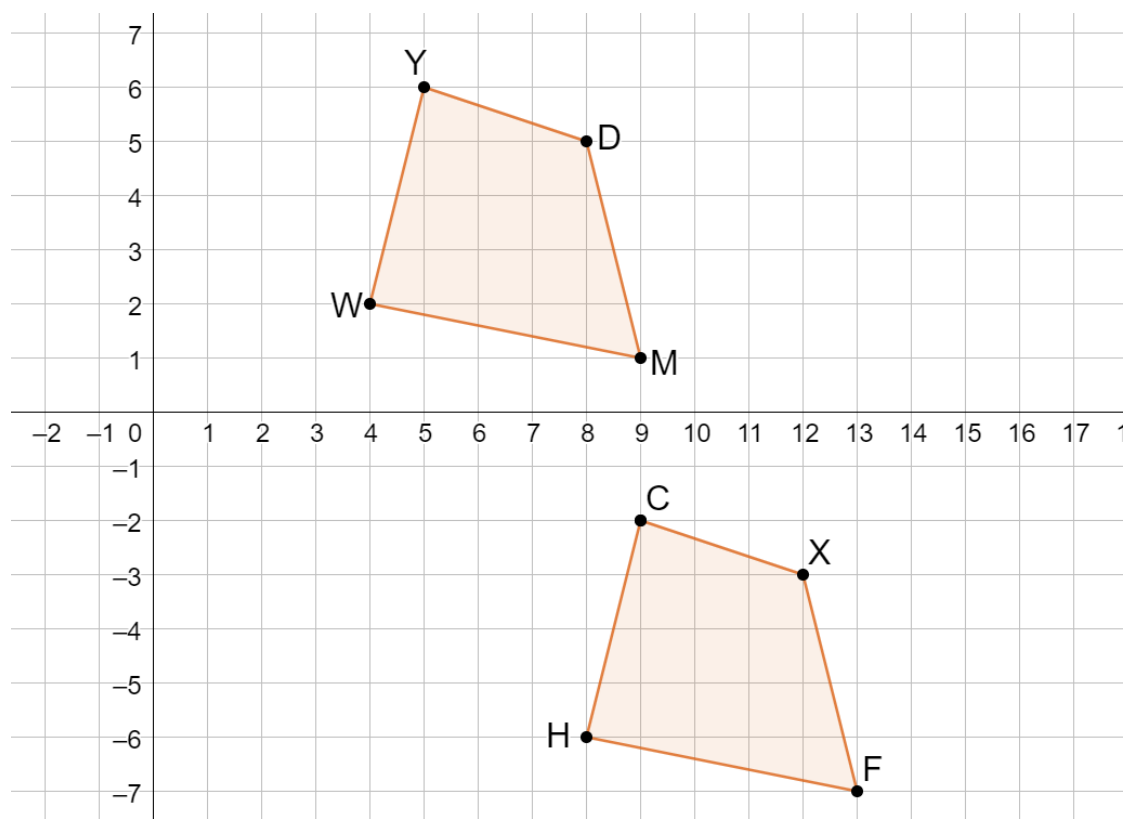
6. Triangles NWG , $N'W'G'$, and $N''W''G''$ are shown on the coordinate grid, where $\Delta N'W'G'$ and $\Delta N''W''G''$ are transformations of ΔNWG .



- Describe the feature(s) of the transformations of ΔNWG that guarantees that each side length of ΔNWG is preserved for this sequence of transformations.
- Determine a sequence of transformations so that when applied, ΔNWG will map onto $\Delta N''W''G''$. Complete the table by writing the algebraic description using coordinates for each mapping.

Mapping	Algebraic Description
$\Delta NWG \rightarrow \Delta N'W'G'$	
$\Delta N'W'G' \rightarrow \Delta N''W''G''$	

7. Quadrilateral $WYDM$ was transformed to create quadrilateral $HCXF$.



- a. Write the algebraic description using coordinates to describe the transformation of quadrilateral $WYDM$ used to create quadrilateral $HCXF$.

Algebraic Description	
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- b. Use the algebraic description and the vertices of quadrilateral $WYDM$ as inputs to show the outputs that result in the vertices of quadrilateral $HCXF$.

Vertex of $WYDM$	Work to determine the x -coordinate of $HCXF$	Work to determine the y -coordinate of $HCXF$
W		
Y		
D		
M		